Vision Agent: A Dual-Encoder Architecture for Disaster Image Classification

DisasterAI — Multi-Agent Disaster Assessment System

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Abstract. We present the Vision Agent, the perceptual module of a multi-agent disaster-assessment pipeline. Given a single image, the system produces a five-way classification (*fire*, *flood*, *normal*, *irrelevant*, *uncertain*), a natural-language description, and an explicit reliability estimate. Classification is performed by a cloud-hosted vision-language model (Qwen, served through Groq, with a local fallback), queried independently on three test-time-augmented views of the input image. Agreement across the three views yields a stability score that is combined with a self-supervised visual-similarity signal from a frozen DINOv2 encoder to produce a final reliability verdict. We describe the architecture, the rationale behind each design choice, and report results from controlled runs on held-out images from the project's fire, flood, and normal classes, each achieving unanimous three-way agreement and a reliable verdict.

1 Introduction

The Vision Agent is one of five cooperating agents in the DisasterAI pipeline (Incident, Vision, Knowledge/RAG, Risk, Report). Its task is narrow but safety-critical: given an image submitted alongside an incident report, decide whether it shows an active fire, an active flood, an unaffected ("normal") scene, content unrelated to disaster assessment (*irrelevant*), or content too ambiguous to classify (*uncertain*), and attach a reliability estimate that downstream agents can condition on. A wrong but confident answer is more harmful to the pipeline than a correct abstention, which motivated most of the design decisions described below.

2 System Architecture

Figure 1 shows the full pipeline. Two branches run on the same input image: a *semantic* branch (test-time augmentation followed by a vision-language model) and a *visual-similarity* branch (a frozen DINOv2 encoder). Their outputs are combined by a Visual Consistency Module and a Reliability Layer before the agent returns a single, schema-validated result.

3 Component Design

3.1 Vision-Language Model: Cloud-First with Local Fallback

The semantic branch is served by a hosted multimodal model (*Qwen*, accessed through the Groq inference API) with a local model (served through Ollama) configured as an offline fallback. Model selection is a single configuration value; no code change is required to switch providers.

Rationale. Three considerations drove this choice over a purely local deployment. First, output quality: in side-by-side comparisons on the project's test images, the cloud-hosted model produced substantially more specific and better-grounded descriptions than the local alternative (Table 1). Second, deployment cost: a cloud API requires no GPU, no model download, and no local inference service, which matters directly for handing the system to a collaborator's machine, where disk space and GPU availability cannot be assumed. Third, resilience: request failures are retried automatically, including honoring the provider's rate-limit back-off interval, before falling back to the secondary provider.

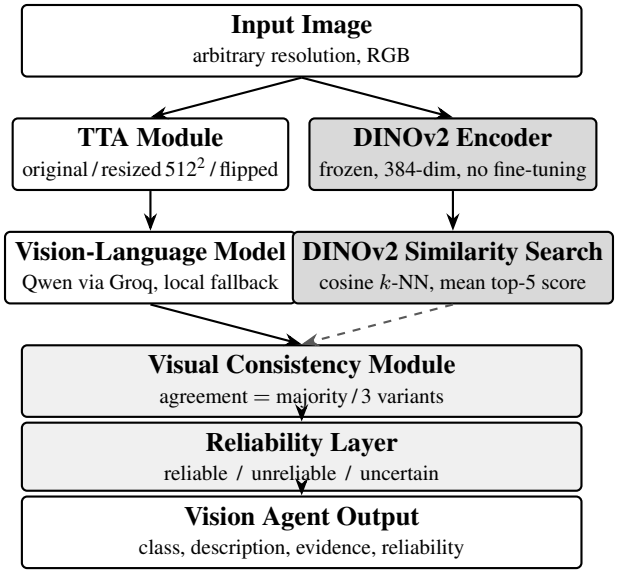


Figure 1: Vision Agent pipeline. White nodes: the semantic (language) branch. Gray nodes: the non-linguistic visual-similarity branch. The dashed arrow denotes a shadow-mode signal: it is logged alongside the reliability verdict but, in the default configuration, does not alter it.

	Local (Ollama)	**Cloud (Qwen via Groq)**
Description	"A wildfire is burning in the mountains."	"Intense orange flames burn along a ridge line, generating a massive plume of thick, dark smoke."
Evidence	generic (flames, smoke, structural collapse)	specific (active flames along a ridge, thick dark smoke plume, glowing embers and fire front)
Deployment	requires local GPU, model download	requires only an API key

Table 1: Output quality and deployment cost, same input image, two VLM backends.

Query design. Each call asks for a single structured JSON object containing the predicted class, a natural-language description, and a short evidence list, in one pass. An earlier iteration issued three separate questions (classification, evidence, description) and occasionally produced internally inconsistent answers — a classification of *fire* accompanying a description of a lake. Merging the three sub-answers into one constrained JSON response removes this failure mode by construction: class and description are now necessarily drawn from the same reasoning pass.

3.2 Five-Way Classification Taxonomy

The agent distinguishes five outcomes, listed in Table 2. The *normal/irrelevant* distinction is deliberate: a street, building, or landscape with no active incident is still within the domain the pipeline is meant to assess (*normal*), whereas a selfie, a document photograph, or an unrelated object is not (*irrelevant*). Separating the two allows the pipeline to reject out-of-domain input early rather than force it into a disaster-relevant answer.

3.3 Test-Time Augmentation

Each image is independently classified in three forms: the original, a version resized to 512×512, and a horizontally mirrored version. Running the same query three times on perturbed inputs exposes a form of prediction instability that a single call cannot reveal. The three transformations were chosen so as not

Class	Definition
fire	active flames, burning structures or vegetation, wildfire smoke columns
flood	water inundation — flooded streets, submerged vehicles or buildings, overflowing rivers
normal	a disaster-relevant scene (street, building, landscape, vehicle, person) with no active incident
irrelevant	content unrelated to disaster assessment entirely — a selfie, a document, a screenshot, an unrelated object
uncertain	the image is too dark, blurred, or ambiguous to classify with confidence

Table 2: Classification taxonomy. The boundary between *normal* and *irrelevant* is established purely through the natural-language class definitions given to the model; no separate rule-based gate exists ahead of the language model call.

to destroy the physical evidence being searched for: a vertical flip would place smoke below ground and water above it; a rotation would change the frame of reference the model is implicitly conditioned on; a colour transform could remove fire's characteristic spectral signature. None of the three is applied.

3.4 DINOv2 Visual Similarity

The second branch extracts a 384-dimensional embedding from a frozen DINOv2 vision transformer (no fine-tuning, no gradient computation) and compares it, by cosine similarity, against a small reference set of previously encoded images. The resulting mean top-5 similarity is a purely visual signal: it contains no natural language and cannot be subject to the same class of hallucination that affects the vision-language model. It is reported as part of the reliability block but, in the default (*shadow*) mode, never changes the predicted class or the reliability verdict on its own; the system is configured to require a corroborating signal (low test-time-augmentation agreement) before the similarity score is allowed to downgrade a result.

3.5 Visual Consistency Module and Reliability Layer

The three test-time-augmentation answers are reduced to an agreement score, the fraction of variants agreeing with the majority class ($1/3$, $2/3$, or $3/3$). An agreement score below a fixed threshold ($2/3$), or the presence of hedging language in the model's own answer, marks the result *unreliable*; in that case, the final predicted class is deliberately replaced with *uncertain* rather than reported as the (unstable) majority vote. This design decision reflects the project's governing principle: a confident wrong answer is more dangerous, in a disaster-assessment context, than an honest abstention.

4 Experimental Results

Three images, one from each of the project's labelled classes (*fire*, *flood*, *normal*), were drawn from the training corpus and passed through the complete pipeline with the cloud vision-language model active. Table 3 summarises the outcome of each run; all three produced unanimous agreement across the three test-time-augmented variants and were marked reliable.

Fire. All three variants converged on the same evidence set: "active flames along a ridge," "thick dark smoke plume," "glowing embers and fire front." The canonical description reads: *"Intense orange flames burn along a ridge line, generating a massive plume of thick, dark smoke."* The five nearest neighbours retrieved by the DINOv2 similarity search were themselves all fire images, corroborating the classification through an entirely independent, non-linguistic signal.

Image	Predicted class	TTA agreement	Reliability	DINOv2 mean sim.
`fire_image0001.jpg`	fire	3/3 (1.00)	reliable	0.859
`flood_image0001.jpg`	flood	3/3 (1.00)	reliable	0.784
`image_5990.png`	normal	3/3 (1.00)	reliable	0.887

Table 3: Results of three controlled runs, one per class, using the cloud vision-language model (Qwen via Groq). TTA agreement is reported as (agreeing variants)/(total variants).

Flood. The model consistently identified "muddy brown water covering the ground," submerged trees with only their canopies visible, and buildings with only their roofs above the waterline, across all three augmented variants, with only minor paraphrasing between runs. Mean DINOv2 similarity (0.784) was lower than for the fire example, consistent with flood scenes exhibiting more visual heterogeneity (aerial versus street-level framing, vegetation density, water turbidity) than the comparatively uniform visual signature of active flame and smoke.

Normal. A photograph of a high-rise building against a sunset sky was correctly classified as *normal* in all three variants, with an empty evidence list in each case — the expected behaviour for this class under the prompt design of Section 3.2, since *normal* and *irrelevant* are defined to carry no fire/flood evidence items. Mean DINOv2 similarity (0.887) was the highest of the three runs; its nearest neighbours were exclusively other *normal*-labelled images.

Scope of this evaluation. These three runs demonstrate correct, stable, and reliable behaviour on held-out examples of each labelled class under the current cloud vision-language backend. They do not constitute a full accuracy benchmark: a quantitative evaluation across a larger, randomly sampled test set, and an explicit test of the *irrelevant* and *uncertain* classes on genuinely out-of-domain images (which the present dataset does not contain), remain open work.

5 Discussion and Limitations

Two limitations are worth stating plainly. First, the *normal/irrelevant* boundary (Section 3.2) has not yet been exercised on a genuinely out-of-domain image — every test to date has used fire, flood, or normal-disaster-scene photographs from the project corpus, none of which approximate a selfie or a document photograph. Second, the DINOv2 reference store used in these experiments is small (built from a limited subset of the project's images); the reported similarity scores are internally consistent but should not be read as calibrated probabilities. Both points are natural next steps rather than defects in the present design.

References

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